# Chapter 5

# Geodesic methods

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*If the data lie on a folded sheet, how do you flatten it back out?*

## 5.1 What this chapter answers

Isomap's premise is this book's premise. Estimate the distances measured *along* the surface, embed those instead of the distances measured *through* the space around it, and a sheet that has been bent without being stretched should come back flat. `R/methods.R` records what every method consumes, and Isomap is the only entry in the registry whose answer to that question is "geodesic distance".

That makes this the one chapter where the method's input and the answer key are the same object. A rigid folding is an intrinsic isometry, so the geodesic distance between two sampled points **is** the Euclidean distance between their rows of `truth`. Not an estimate of it, and not a limit approached as the sample grows: the exact number, in closed form, before any graph is built.

The consequence is that Isomap's error decomposes into two terms that can be measured apart rather than argued about. A **graph term**, from approximating geodesics by shortest paths on a $k$-nearest-neighbour graph. An **embedding term**, from putting the resulting distance matrix into two dimensions with classical scaling. Hand `cmdscale()` the exact geodesics and the second term vanishes to floating point. Everything that is left belongs to the first.

This chapter measures that first term across the fold sweep, predicts where it turns from a bias into a break, and reports what the break actually is on this object — which is not the mechanism the plan of record predicted.

Diffusion maps is here as the contrast. It sits in the same registry family and consumes *ambient* distance, and the gap between the two curves is the value of consuming the right thing.

## 5.2 Setup

```
grid   <- read_run("data/processed/benchmark-grid.rds",
 ↪  "2483a468d82d4d64")
sweeps <- read_run("data/processed/part2-sweeps.rds",
 ↪  "16d43ff4992c0e58")
p <- miura_ori(nx = 6L, ny = 6L, alpha = pi / 3)
s <- sample_manifold(p, theta = 0.5, n = N_DEFAULT, seed =
 ↪  BENCH_SEEDS[1])
```

The main grid fixes $k$ at `K_DEFAULT`; `part2-sweeps.rds` is the artefact that moves it, over 7 values at 3 fold angles and all 20 benchmark seeds. Chapter 4 explains what `read_run()` refuses.

## 5.3 Background

### 5.3.1 Isomap, in three stages

Isomap (Tenenbaum et al., 2000) is classical multidimensional scaling (Torgerson, 1952) with the distance matrix replaced. Build a $k$-nearest-neighbour graph on the ambient points, weight each edge by its ambient length, take all-pairs shortest paths, and hand the result to `cmdscale()`. The first two stages estimate the geodesic metric; the third is the same eigenproblem Chapter 4 already ran. Read as a kernel method (Schölkopf et al., 1998; Lee and Verleysen, 2007) it is classical scaling with a geodesic kernel, which is exactly how Chapter 4 hands the reader over.

The estimate is justified asymptotically. Graph shortest paths converge to true geodesics as the sample density rises, at a rate controlled by the sampling radius and the surface's curvature (Bernstein et al., 2000). Both stages of that guarantee are worth naming here, because both fail on a crease pattern: the convergence is stated for a manifold of positive reach, and a crease has reach exactly zero (Chapter 2), and it is asymptotic in a density this book fixes at `N_DEFAULT` points.

```
reg <- method_table()
fam <- reg[reg$family == "geodesic", c("method", "label",
 ↪  "consumes", "stochastic")]
knitr::kable(fam, row.names = FALSE,
             col.names = c("registry name", "label", "consumes",
              ↪  "stochastic"),
             caption = "The geodesic family of the embedding
              ↪  registry.")
```

Table 5.1 is the chapter in one row each. Isomap consumes geodesic distance and is deterministic. Diffusion maps consumes **ambient** distance and is stochastic, and both of those are easy to get wrong.

Table 5.1: The geodesic family of the embedding registry.

| registry name | label | consumes | stochastic |
|---|---|---|---|
| isomap | Isomap | geodesic | FALSE |
| diffusion | Diffusion map | ambient | TRUE |

### 5.3.2 Diffusion maps, and why it is filed here

Diffusion maps (Coifman and Lafon, 2006) puts a Gaussian kernel on the ambient distances, normalises it into a Markov chain, and embeds the points so that Euclidean distance in the embedding approximates *diffusion distance* — the difference between the transition distributions of a random walk started at each point. Diffusion distance is a geodesic-flavoured quantity, which is why the family label is what it is, but nothing in the construction estimates a geodesic. The kernel sees ambient distance and only ambient distance, and the registry's `consumes` field says so. The same holds for Laplacian eigenmaps (Belkin and Niyogi, 2003), which Chapter 6 takes up.

Two implementation facts travel with it. It is **stochastic** — `diffusionMap::diffuse()` uses a randomised eigensolver — so `embed()` refuses to run it without a seed. And its bandwidth is set from the $k$-th nearest-neighbour distance rather than the median pairwise distance, because the outlier noise model inflates the second and not the first; `R/methods.R` records what that cost when it was done the obvious way.

> **Two changes from the plan of record.** The outline for this chapter named `igraph::distances()` for the shortest paths and locally linear embedding as a third method. Neither survived. `igraph` is not installable in this project — a missing system library — so `R/metrics.R` computes all-pairs shortest paths by min-plus relaxation with a work list, which is Bellman–Ford rather than Dijkstra [@dijkstra1959note] and was timed against two alternatives at the book's sample size before it was kept. LLE consumes neighbourhood structure, not geodesics, and is in the neighbor embeddings chapter with the other methods that do.

## 5.4 The geodesic you do not have to estimate

### 5.4.1 The claim, and its one condition

Folding preserves distance along the surface exactly (Chapter 2). So the length of a path on the folded sheet equals the length of its preimage in the flat chart, and the shortest path in the chart between two points is the straight segment between them. Therefore `dist(s$truth)` is the geodesic distance matrix — not a discretisation of it.

The condition is that the straight segment stays on the paper. A Miura's unfolded outline is not convex, so this is a real condition and not a formality; `sample_manifold()`'s default `boundary = FALSE` trims the sample to a convex sub-rectangle, and the trimming is what makes the equality hold rather than a

bound.

```
dU   <- reference_dist(s, "chart")
dA   <- reference_dist(s, "ambient")
same <- outer(s$facet, s$facet, "==")
up   <- upper.tri(dU)

within  <- max(abs(dA - dU)[same & up])          # isometry,
 ↪  inside a facet
across  <- max((dA - dU)[!same & up])            # contraction,
 ↪  across a crease
exit    <- chart_exit_fraction(s, p)             # segments that
 ↪  leave the sheet
```

Three checks, and each one is a different half of the claim. Inside a facet ambient and chart distance agree to 1.1e-15, which is rounding on a closed form. Across a crease the ambient distance is strictly smaller — the largest signed difference over all 308,887 such pairs is -6.2e-07, a negative number. And `chart_exit_fraction()` reports that 0 of sampled segments leave the sheet, so the chart distance is the geodesic for every pair in this sample and not merely a lower bound on it.

### 5.4.2 The decomposition

```
Y_exact <- cmdscale(as.dist(dU), k = EMBED_DIM)           # exact
 ↪  geodesics in
dG      <- reference_dist(s, "graph", k = K_DEFAULT)      # what
 ↪  Isomap estimates
Y_iso   <- embed("isomap", s, k = K_DEFAULT)

err_embed <- reconstruction_error(Y_exact, s$truth)
err_total <- reconstruction_error(Y_iso,   s$truth)
agree     <- max(abs(unname(Y_iso) - unname(cmdscale(as.dist(dG),
 ↪  k = EMBED_DIM))))
graph_dev <- sqrt(mean(((dG - dU)[up])^2)) /
 ↪  sqrt(mean((dU[up])^2))
```

Feed classical scaling the exact geodesic distances and it returns the chart, at a Procrustes error of 5.7e-16. That is the embedding term, and it is zero: the chart is a two-dimensional Euclidean configuration, so double-centring its squared distances and taking two eigenvectors inverts the construction exactly.

Isomap on the same sample scores 0.0266. It differs from `cmdscale()` on the graph distances by 7.9e-15, because it *is* `cmdscale()` on the graph distances. So the whole of that 0.0266 is the graph term, and the graph term is small in the distances themselves — the shortest-path matrix departs from the exact geodesics by 2.95 per cent in relative root-mean-square.

This is the sense in which the chapter's title is not a metaphor. Isomap's error on a crease pattern has one cause, it is a property of a graph rather than of an optimiser, and it can be looked at directly.

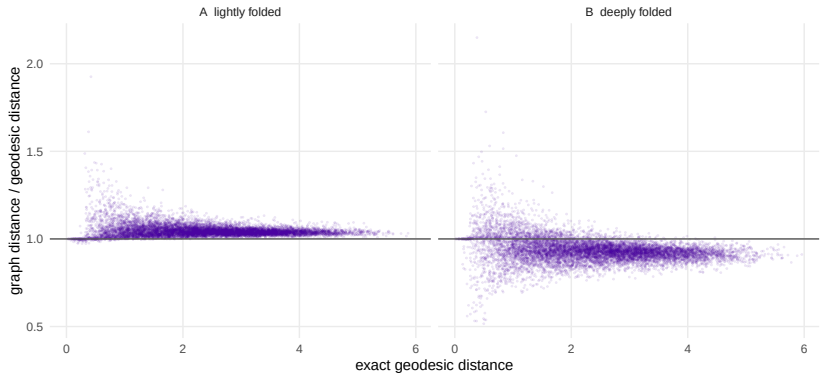



Figure 5.1: Shortest-path distance on the ambient neighbourhood graph against the exact geodesic distance, as a ratio, for a lightly folded sheet and a deeply folded one. The horizontal line at one is exactness. Points above it are the discretisation bias of walking along graph edges; points below it are distances the fold has contracted. A random ten thousand pairs are drawn in each panel.

Figure 5.1 is the graph term seen twice. At a small fold angle it is a *positive* bias: a path made of graph edges zigzags where a geodesic runs straight, so shortest paths come out too long, and the scatter collapses onto a tight band just above one as the pair gets further apart and the zigzag averages out. At a large fold angle that band has dropped below the line. Edges that cross a crease are shortened by the $\sin(\rho/2)$ factor Chapter 2 derives, shortest paths route through as many creases as they can, and the estimate falls below the truth. The two biases are different mechanisms with opposite signs, and `geo-diagnostics` is where they cancel.

## 5.5 Results

```
clean <- subset(grid, noise == "none" & ran)
cur   <- aggregate(cbind(rmse, qnx) ~ method + theta, clean,
 ↪  mean)
band  <- aggregate(rmse ~ method + theta, clean,
                   function(x) c(lo = min(x), hi = max(x)))
band  <- data.frame(band[, c("method", "theta")], band$rmse)
cur   <- merge(cur, band, by = c("method", "theta"))
cur   <- subset(cur, method %in% c("isomap", "diffusion", "pca"))

at <- function(m, t, col = "rmse") cur[[col]][cur$method == m &
 ↪  cur$theta == t]
wide <- reshape(cur[, c("theta", "method", "rmse", "qnx")],
                idvar = "theta", timevar = "method", direction =
                 ↪  "wide")
wide <- wide[order(wide$theta), ]
cross_rmse <- wide$theta[which(wide$rmse.isomap <
 ↪  wide$rmse.pca)[1]]
```

```
cross_qnx  <- wide$theta[which(wide$qnx.isomap  >
 ↪  wide$qnx.pca)[1]]
noisy <- aggregate(rmse ~ noise + theta, subset(grid, method ==
 ↪  "isomap" & ran),
                   mean)
```

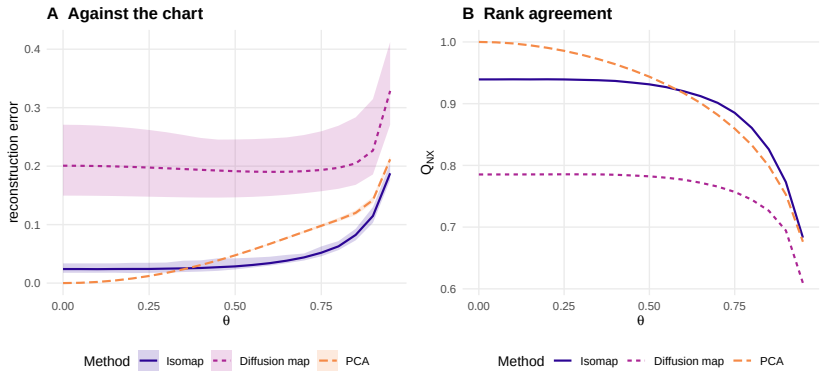



Figure 5.2: Isomap and diffusion maps over the fold sweep, with PCA from Chapter 4 as the reference. Panel A is reconstruction error against the chart, panel B is $Q_{NX}$ at $K_{QNX}$ against the chart. Lines are means over the twenty benchmark seeds and bands in panel A are their full range. Lower is better in A, higher in B.

Three readings of Figure 5.2, in order of how much they cost the reader to accept.

**Isomap wins, eventually, and pays an entry fee for it.** On the flat sheet its error is 0.024 where PCA's is zero: the graph term is there at $\theta = 0$, because a neighbourhood graph is a discretisation whether or not anything has been folded. Isomap does not overtake PCA on reconstruction error until $\theta = 0.4$, and on $Q_{NX}$ not until $\theta = 0.6$. From 0.4 to the end of the sweep it is the best method in the book on this grid, and at the last fold angle it scores 0.188 against PCA's 0.212.

That the two metrics put the crossing in different places is not a discrepancy to be resolved. Reconstruction error reads distances and $Q_{NX}$ reads rank order, and Chapter 9 is where the disagreement is the subject rather than a nuisance.

**The curve has a shape, and the shape is the point.** Isomap's error is essentially flat through the first half of the sweep — it grows by a factor of 1.19 between the flat sheet and the middle — and then rises by a factor of 6.58 over the second half. A benchmark reporting one fold angle would have called this method robust or fragile depending entirely on which angle it chose.

**Diffusion maps does not improve on PCA anywhere.** Its error is close to flat across the whole sweep, from 0.201 to 0.329, and it is the *worst* of the three at 20 of the 20 fold angles in the sweep. A method that consumes ambient distance and embeds a random walk on it has no mechanism for undoing a fold; it is in this chapter to show that the family label is not what does the work.

Noise moves Isomap's curve without changing it. At $\theta = 0$ the error is already 0.124 under ambient Gaussian noise and 0.416 under outliers, on a sheet that is not folded at all. Displaced points are new neighbours, and a graph method's whole input is who is a neighbour.

## 5.6 When does the graph bridge?

A **fold jump** is a $k$-nearest-neighbour edge between two points the sheet holds far apart — the classical short circuit that destroys Isomap on tightly wound surfaces (Balasubramanian and Schwartz, 2002). It is the failure this chapter was specified to predict, and the prediction can be made without running anything.

Two quantities cross. `facet_gap(p, theta)` is the exact minimum ambient distance between two facets that share no vertex: a property of the surface alone, no sample and no density (Chapter 2 uses it to check that the sheet never passes through itself). Against it stands the $k$-nearest-neighbour radius $r_k$ in the chart. That radius is a constant of the sweep, not a function of $\theta$: `sample_manifold()` draws the chart points first and folds them after, so for a fixed seed `s$truth` is the *same matrix* at every fold angle. The graph acquires its first fold jump when the surface's own clearance falls below the radius at which the sample connects points.

```
KS   <- c(10L, 20L, 30L, 45L)
s0   <- sample_manifold(p, theta = 0, n = N_DEFAULT, seed =
 ↪  BENCH_SEEDS[1])
dU0  <- as.matrix(dist(s0$truth))
Lfar <- 0.25 * max(dU0)                          # branch_gap()'s
 ↪  "far apart"
d0   <- dU0; diag(d0) <- Inf
r_k  <- vapply(KS, function(k)
  median(apply(d0, 1L, function(r) sort.int(r, partial = k)[k])),
   ↪  numeric(1))

th_fine <- seq(0.70, 0.99, by = 0.01)            # deliberately past
 ↪  THETA_GRID
gap     <- vapply(th_fine, function(t) facet_gap(p, t)$gap,
 ↪  numeric(1))
gapfun  <- stats::approxfun(th_fine, gap)
predicted <- vapply(r_k, function(r) tryCatch(
  stats::uniroot(function(t) gapfun(t) - r, range(th_fine))$root,
  error = function(e) NA_real_), numeric(1))

jumps <- function(theta, seed) {
  st <- sample_manifold(p, theta = theta, n = N_DEFAULT, seed =
 ↪  seed)
  nn <- FNN::get.knn(st$X, k = max(KS))$nn.index
  du <- as.matrix(dist(st$truth))
  vapply(KS, function(k) sum(du[cbind(rep(seq_len(nrow(nn)),
   ↪  times = k),
                                      as.vector(nn[,
                                       ↪  seq_len(k)]))] > Lfar),
```

Table 5.2: Predicted and measured onset of the first fold jump.

| k | chart radius | predicted onset | measured onset | edges at flat |
|---|---|---|---|---|
| 10 | 0.292 | 0.974 | 0.971 | 0 |
| 20 | 0.413 | 0.945 | 0.944 | 0 |
| 30 | 0.507 | 0.909 | 0.914 | 0 |
| 45 | 0.629 | 0.832 | 0.855 | 0 |

```
        numeric(1))
}
counts <- t(vapply(th_fine, function(t)
  rowMeans(vapply(BENCH_SEEDS, function(sd) jumps(t, sd),
    ↪  numeric(length(KS)))),
  numeric(length(KS))))
control <- rowMeans(vapply(BENCH_SEEDS, function(sd) jumps(0,
 ↪  sd),
                           numeric(length(KS))))
measured <- vapply(seq_along(KS), function(a) {
  i <- which(counts[, a] >= 1)[1]
  if (is.na(i)) NA_real_
  else if (i == 1L) th_fine[1]
  else stats::approx(counts[(i - 1):i, a], th_fine[(i - 1):i],
    ↪  xout = 1)$y
}, numeric(1))
onset <- data.frame(k = KS, r_k = r_k, predicted = predicted,
                    measured = measured, at_flat = control)
```

An edge counts as a fold jump when its endpoints are more than `Lfar` apart in the chart, which is `branch_gap()`'s own definition of "far apart along the surface" reused so that the chapter and the difficulty statistic mean the same thing. The onset is where the *expected* number of such edges reaches one, because the first jump is a tail event and one edge in one seed is not an onset.

The fold sweep here runs past `max(THETA_GRID)`, deliberately and uniquely in this book. Whether the short circuit arrives before or after the benchmark's own ceiling is exactly the question, so the measurement may not be confined to the answer.

The criterion calibrates itself. At $\theta = 0$ the sheet is flat, ambient and chart distance coincide, and any edge counted there is sparse sampling rather than folding. That control is 0 at every $k$ tested, so the counts below have no false positives in them.

Table 5.2 is the chapter's headline promise, kept. The analytic onset and the measured one agree to 0.023 in $\theta$ at worst — under half a step of the benchmark's own grid — over four neighbourhood sizes spanning a factor of 4.5. The residual has the sign it should: `facet_gap()` is a minimum over the whole of two facets, and a finite sample does not put points exactly at the closest-approach spot, so the measurement is late.

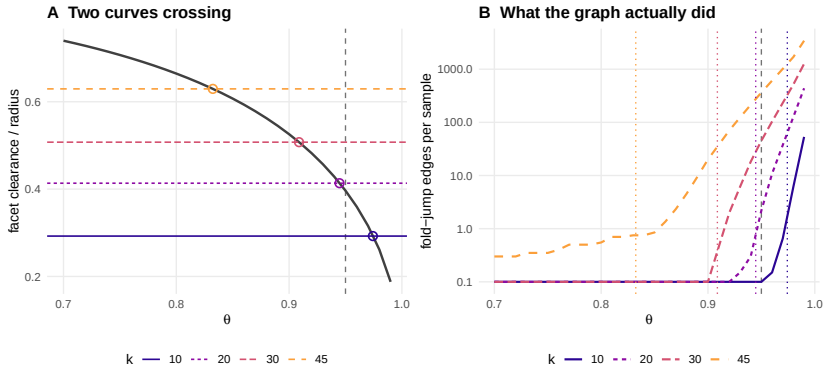



Figure 5.3: The analytic onset against the measured one, over a fold sweep deliberately extended past the benchmark's own ceiling — the vertical dashed line at the largest fold angle the grid uses. Panel A is the exact clearance between non-touching facets as the sheet closes, with the chart $k$-nearest-neighbour radius for four neighbourhood sizes drawn as horizontal lines; each crossing is a predicted onset, marked with an open circle. Panel B is the measured number of fold-jump edges per sample, averaged over the twenty benchmark seeds, with the same predictions marked on the horizontal axis.

The row that matters is the first one, and it is a finding about the benchmark rather than about Isomap.

> **The committed grid never enters the short-circuit regime.** At $k$ = `K_DEFAULT` — the neighbourhood size every cell of `benchmark-grid.rds` runs at — the predicted onset lies past `max(THETA_GRID)`, and the measured count inside the sweep never leaves zero. The classical failure that geodesic methods are known for is not what this benchmark, as committed, measures.

The count inside the committed sweep is 0 at every fold angle in it, over all 20 seeds. The two boundaries are independent, which is what makes that worth saying rather than assuming. `THETA_GRID` stops where it does because at $\theta = 1$ the sheet has no extent in one direction and every method would be embedding a collapsed object — a reason with nothing to do with neighbourhood graphs (`R/constants.R` states it). The short circuit at $k$ = `K_DEFAULT` arrives at 0.974 predicted and 0.971 measured, which is 0.021 past that ceiling. Nothing was arranged; the grid simply stops just short.

So everything in Figure 5.2 — the whole of Isomap's degradation from 0.024 to 0.188 across the sweep — happened without a single fold jump. Raising $k$ is what crosses the boundary inside the grid, and that is the subject of the next section.

## 5.7 Diagnostics

If the short circuit is absent, what is doing the damage? The $k$ sweep answers it, and the answer is that a neighbourhood size is not a setting to be tuned once.

```
sw   <- aggregate(cbind(rmse, qnx) ~ method + k_tck + theta,
                  subset(sweeps, ran), mean)
best <- do.call(rbind, lapply(split(subset(sw, method ==
 ↪  "isomap"),
                                    subset(sw, method ==
                                     ↪  "isomap")$theta),
                               function(d) d[which.min(d$rmse),
                                ↪  ]))
dif  <- subset(sw, method == "diffusion")
iso  <- subset(sw, method == "isomap")
ths  <- sort(unique(sweeps$theta))
kmax <- max(sweeps$k_tck)
kbest <- function(t) best$k_tck[best$theta == t]
err   <- function(t, k) iso$rmse[iso$theta == t & iso$k_tck == k]
carry <- function(from, to) err(to, kbest(from)) / err(to,
 ↪  kbest(to))
```

The neighbourhood size that minimises Isomap's reconstruction error falls as the sheet closes: 45 at $\theta = 0.2$, 20 at 0.5, and 12 at 0.8. Neither end's choice survives the other. Carrying the shallow optimum to the deepest fold multiplies the error by 1.51; carrying the deep optimum back to the shallowest multiplies it by 3.81. There is no one $k$ for this pattern, and the relative cost of the wrong one is larger at the shallow end — where the absolute error is smaller by an order of magnitude, so the two ends are not simply harder and easier versions of one problem.

```
KB   <- c(5L, 10L, 20L, 45L)
bias <- do.call(rbind, lapply(ths, function(t) {
  st <- sample_manifold(p, theta = t, n = N_DEFAULT, seed =
 ↪  BENCH_SEEDS[1])
  du <- reference_dist(st, "chart"); u <- upper.tri(du)
  do.call(rbind, lapply(KB, function(k)
    data.frame(theta = t, k = k,
               bias = median(((reference_dist(st, "graph", k = k)
                ↪  - du) / du)[u])))))
}))
zero <- vapply(split(bias, bias$theta), function(d)
  tryCatch(exp(stats::approx(d$bias, log(d$k), xout = 0)$y),
   ↪  error = function(e) NA_real_),
  numeric(1))
```

Figure 5.4 puts the mechanism next to its consequence. Panel B is the two biases of Figure 5.1 as functions of $k$. Raising $k$ shortens the zigzag and kills the positive bias; it also lengthens every edge, and a longer edge crosses more creases and is contracted by more of them. The crossing point — where the median shortest path is neither long nor short — falls from $k \approx 35$ at $\theta = 0.2$ to 14 at 0.5 and 6.8 at 0.8, in the same order and over nearly the same range as the error minima in panel A. The two do not coincide, and should not: panel A minimises a Procrustes fit that pays for variance as well as bias.

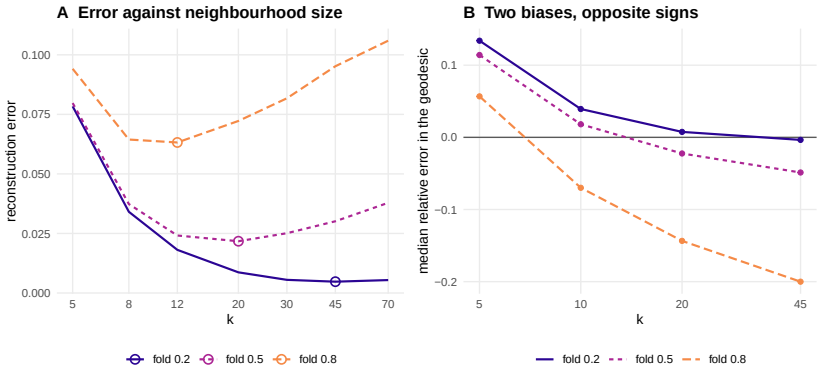



Figure 5.4: Why the optimum moves. Panel A is Isomap's reconstruction error against neighbourhood size at three fold angles, from the committed sweep; the open circles mark each curve's minimum. Panel B is the median relative departure of the shortest-path distance from the exact geodesic, measured at the same three fold angles: positive is the discretisation bias of walking along edges, negative is the fold contracting them, and each curve crosses zero at a smaller neighbourhood size than the last.

Diffusion maps has no such trade-off. Its error falls monotonically in $k$ at all three fold angles, from 0.208 at $k = 5$ to 0.132 at $k = 70$ in the middle of the sweep, because $k$ there sets a kernel bandwidth and not a graph. It never approaches Isomap's optimum, and its best setting is the largest one offered — a curve still falling at the edge of the sweep is a sweep that stopped too early, and this one is reported as such.

Two housekeeping facts from the artefacts, both of which would have been warnings scrolling past in a long run. Isomap's `k_effective` column equals the requested $k$ in every one of the 1200 main-grid rows and every sweep row, so the neighbourhood graph never disconnected and never needed repairing. Diffusion maps needed a widened kernel in 399 of 1200 cells and declined outright in 1; those are recorded in the `tuning` and `status` columns rather than folded into the mean.

## 5.8 Where it fails

**The predicted failure is not the observed one.** This chapter was specified to pair an analytic short-circuit onset with a measured one, and it does — to 0.023 in $\theta$. What the specification also assumed, and what is false, is that the short circuit is what breaks Isomap on this object. At the grid's neighbourhood size it never occurs inside the sweep, and Isomap's error still rises by a factor of 7.82 across it. The mechanism that actually operates is edge contraction: present at every $\theta > 0$, graded rather than catastrophic, and invisible to any diagnostic built to look for a bridge. A reader who arrives expecting Isomap to collapse here will not see it collapse — it is the best method on this grid above $\theta = 0.4$ — and the reason is that the benchmark's difficulty axis stops short of the regime that would do it.

**When the bridge does form, it is not graceful.** Shortest path takes a *minimum* over routes, so a single bad edge is not averaged against the good ones — it is preferred by every pair whose route it shortens.

```
z   <- pair[[2]]                                  # the deeply
 ↪  folded sample
dAf <- as.matrix(dist(z$s$X))
adj <- .knn_graph(dAf, K_DEFAULT)                 #
 ↪  reference_dist()'s internals,
G0  <- .apsp_minplus(adj, dAf)                    #   reached
 ↪  directly so that the
far <- as.matrix(dist(z$s$truth)) > Lfar          #   graph can
 ↪  be edited between

worst <- dAf; worst[!far] <- Inf
ij  <- arrayInd(which.min(worst), dim(worst))
edges <- sum(adj) / 2
adj[ij[1], ij[2]] <- adj[ij[2], ij[1]] <- TRUE
G1  <- .apsp_minplus(adj, dAf)

u2      <- upper.tri(G0)
touched <- mean((G1 < G0 - TOL$default)[u2])
worst_p <- max(((G0 - G1) / G0)[u2])
before  <- reconstruction_error(cmdscale(as.dist(G0), k =
 ↪  EMBED_DIM), z$s$truth)
after   <- reconstruction_error(cmdscale(as.dist(G1), k =
 ↪  EMBED_DIM), z$s$truth)
```

One edge is added to a graph that already has 4,692 of them — the single closest ambient approach between two points the chart holds far apart, which is exactly the pair `branch_gap()` reports. It shortens the shortest path for 5.15 per cent of all pairs, by up to 38.9 per cent, and raises the reconstruction error from 0.0621 to 0.0767. Nothing about the sample changed; one entry of an adjacency matrix did.

Figure 5.5 is what those errors look like. Neither panel is a failure a reader would spot by eye, which is the argument for having an answer key: the deeply folded embedding is 2.6 times worse than the lightly folded one and looks like a slightly wobblier version of it.

**The estimate has no guarantee here.** The convergence of graph distances to geodesics is stated for a manifold of positive reach and a density that grows (Bernstein et al., 2000; Niyogi et al., 2008). A crease has reach zero and this book fixes $n$ at `N_DEFAULT`. What replaces the guarantee is the measurement above, and that is a weaker thing than a theorem.

**One family, one size, one density.** Every onset in Table 5.2 is a statement about a six-by-six Miura at `N_DEFAULT` points. The radius $r_k$ falls as $\sqrt{k/n}$, so the same prediction at four times the sample size moves every onset later, and the sweep that would establish that has not been run.

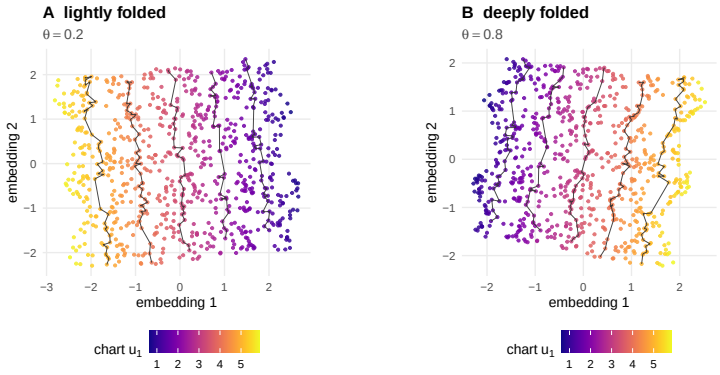



Figure 5.5: What Isomap returns, coloured by the exact chart coordinate along the corrugation, with grey level sets of that coordinate drawn over the points. Panel A is the lightly folded sheet and panel B the deeply folded one. Both recover the chart; the damage in B is a crowding of the level sets, not a tear.

## 5.9 Reproduce this

Both artefacts are committed and neither is generated at render time. `scripts/run-benchmark-grid.R` produces `benchmark-grid.rds` and `scripts/run-part2-sweeps.R` produces `part2-sweeps.rds`; each declares its artefact path in its header, and `scripts/check-artefact-producers.R` checks that every artefact the book specifies has one. The digests in `geo-setup` are what stop these sentences from being read against a different run.

The claims the chapter rests on are assertions in the test suite:

- `tests/testthat/test-folding.R` — facet isometry across the sweep, and `facet_gap()` against a brute-force point sample.
- `tests/testthat/test-metrics.R` — `reference_dist()`'s three geometries, the neighbourhood graph's connectivity repair, and the shortest paths against a reference implementation.
- `tests/testthat/test-methods.R` — the registry contract, including that a stochastic method called without a seed is an error.

Run them with `Rscript tests/testthat.R` from the repository root.

## 5.10 Further reading

Tenenbaum, de Silva and Langford (Tenenbaum et al., 2000) is Isomap's source and still the clearest statement of the three-stage construction. The convergence of graph distances to geodesics is Bernstein and co-authors (Bernstein et al., 2000); read it together with Balasubramanian and Schwartz's comment (Balasubramanian and Schwartz, 2002), which is where the topological instability this chapter measures was first named, and with Niyogi, Smale and Weinberger (Niyogi et al., 2008) for what a sampling guarantee needs in order to exist.

Coifman and Lafon (Coifman and Lafon, 2006) introduce diffusion maps and the diffusion distance the embedding approximates; Belkin and Niyogi (Belkin and Niyogi, 2003) is the closely related Laplacian construction Chapter 6 takes up. Lee and Verleysen (Lee and Verleysen, 2007) set out the whole family as one kernel eigenproblem, which is the framing that makes classical scaling (Torgerson, 1952), kernel PCA (Schölkopf et al., 1998) and Isomap three settings of one method rather than three methods. Dijkstra (Dijkstra, 1959) is where shortest paths on a weighted graph begin, though `R/metrics.R` does not use his algorithm and says why.